
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

Anonymous Author(s)

Affiliation

Address

email

Abstract

Despite their impressive and increasingly general-purpose performance, the broad family of generative image models trained on large-scale datasets frequently fail to produce images with seemingly simple concepts—e.g., human hands or objects appearing in groups of four—that are reasonably expected to appear across the training data. Across model families, these failure modes have largely been documented anecdotally, leaving open the more general question of whether they reflect idiosyncratic, per-model anomalies or more structural limitations shared broadly by such models. To address this question at scale, we introduce a general and systematic approach for identifying and characterizing “conceptual blindspots”—concepts present in the training data but absent or misrepresented in a model’s generations. Our method leverages sparse autoencoders (SAEs) to extract interpretable concept embeddings, enabling a quantitative and model-agnostic comparison of concept prevalence between real and generated images. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—enabling fine-grained analysis of conceptual disparities across a wide range of concepts. Instantiated on four widely used generative models spanning distinct architectures and training pipelines (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), our approach reveals specific suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual datapoint level, the same framework further isolates memorization artifacts — instances where models reproduce highly specific visual templates seen during training. Overall, we propose a theoretically grounded and broadly applicable framework for systematically identifying conceptual blindspots in generative models by assessing their conceptual fidelity with respect to the underlying data-generating process.

1 Introduction

Generative image models trained on large scale datasets have achieved unprecedented and increasingly general capabilities, enabling their deployment across a wide range of applications both within the vision domain [1–9] and in many settings well beyond it [9–15]. Despite this broad success, a growing body of qualitative [16–18] and quantitative studies [19, 20] has shown that, across diverse models and settings, generation can struggle with even relatively simple concepts, e.g., human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. In fact, when prompted to generate images containing such concepts, models across families tend to produce outputs with related structures, but not precisely the ground truth concept—e.g., producing hands with six fingers. These failure modes, which we call “conceptual blindspots”¹, can be unintuitive, since one may reasonably expect that models trained at scale have

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: $\delta(k)$ quantifies a model’s tendency to over- or under-generate a concept c_k compared to its natural-data frequency, providing a general, per-concept diagnostic that applies uniformly across models. We deem concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. The depicted images, generated by four widely used generative image models spanning distinct architectures, illustrate representative images exhibiting conceptual blindspots as well as aligned concepts.

had ample exposure to demonstrations accurately detailing such concepts. This raises a broader question: whether such failures reflect intriguing quirks of a few specific concepts, or whether they are a manifestation of a more systematic and general phenomenon under which, across a broad spectrum of concepts, models are consistently unlikely—or overly likely—to produce images containing them.

Answering this question at the level of generality it demands remains infeasible with existing approaches for evaluating generative image models [28, 29]. Specifically, existing approaches generally rely on coarse-grained measures that are meant to assess image realism, e.g., FID [30], and hence do not capture the full range of distributional failures. Methods like CLIPScore evaluate generation diversity or distribution-coverage statistics [31–33], offering partial insights into our question, but not at the granularity of fine-grained features or concepts [34, 35], making it difficult to identify conceptual blindspots. Finally, qualitative analyses for evaluating generative models, such as participant surveys [5, 17, 36–38] or open-ended exploration [39], can identify failures in a model’s ability to capture certain concepts, but do not offer a scalable, general-purpose methodology that can be readily repeated across model families and reused for their systematic comparison.

This work. Motivated by the above, we argue that identifying and analyzing conceptual blindspots across generative image models requires a methodology that, in an automated and unsupervised manner, can elicit any concept in the data distribution that exhibits a mismatch between its odds of generation under the true data-generating process versus the trained model—and that does so in a way that transfers across model families and concept types. Our contributions in this work are three-fold:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We introduce a general, systematic framework for identifying and quantifying conceptual blindspots in generative image models relative to natural images (Section 3). This formalization moves beyond anecdotal or human-defined concept evaluations, offering a principled and broadly applicable approach for characterizing model limitations.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We develop a structured, model-agnostic methodology using sparse autoencoders (SAEs) to extract and compare concept distributions between natural and synthesized images (Section 4). To this end, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32, 000 concepts, the largest such RA-SAE to date.
- **Exploratory Tool and Analysis.** Our exploratory web tool enables both distribution- and datapoint-level analysis of blindspots across models (Sections 5.1–5.4). As a representative instantiation spanning distinct architectures, we apply our method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5). We identify specific instances of both suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), shown in Fig. 1.

The model, code, and web tool are available at a software repository.

74 2 Related Work

75 2.1 Explainability in Vision

76 Early work in explainable AI, spanning computer vision and neural networks more broadly, focused
77 on methods for attributing predictions to influential input regions [40–42]. However, across settings
78 these methods offered limited semantic information about learned representations and often produced
79 incorrect explanations [43–45]. To address these issues, concept-based interpretability [46] emerged
80 as a general strategy to identify semantically meaningful directions in neural networks, revealing not
81 just where they look but what concepts and structures they employ [47–49].

82 Recent work demonstrates that popular concept-based interpretability methods—ACE [44],
83 CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—can be viewed uniformly as instances
84 of the same dictionary learning task under different constraints [52]. Among these approaches, sparse
85 autoencoders (SAEs) have emerged as particularly scalable and broadly applicable for concept-
86 based interpretability. While recent studies reveal some limitations of the original SAEs—including
87 overly specific features [53], compositionality challenges [54], and limited intervention effects [55]—
88 improved SAE variants have emerged, including archetypal SAE (RA-SAE) [56].

89 Beyond SAEs, other interpretability methods—including prompt-based probing [57], attention map
90 or activation visualizations [58, 59], and dataset-level statistics [32, 33] (e.g., diversity or distribution
91 coverage metrics)—offer only partial insights toward answering these questions. Crucially, across
92 these approaches, granularity is limited: they focus on full images or prompts instead of fine-grained
93 features and concepts [34, 35]. Furthermore, they generally depend on subjective interpretation and
94 do not distinguish between the various failure modes that can arise [60]. Existing methods and metrics
95 are hence broadly inadequate for systematically identifying feature- and concept-level weaknesses of
96 generative image models [28].

97 2.2 Generative Image Models

98 Diffusion-based methods have become dominant across a broad spectrum of modalities in generative
99 vision modeling, including image [4, 61–64], video [65–72], and 3D [7, 73–75]. In the domain
100 of image generation, this lineage can be traced back to denoising diffusion probabilistic models
101 (DDPMs) [76], which were later generalized to non-Markov diffusion processes with denoising
102 diffusion implicit models (DDIMs) [77].

103 The Stable Diffusion (SD) [9] model family made DDPMs broadly accessible across both the
104 research and open-source communities. The original SD was followed by several subsequent versions,
105 including SD 2 [78], SD 3 [79], and SD XL [80]. A wide range of modifications and extensions
106 of the SD architecture have since emerged, enabling additional constraints on the diffusion process
107 (e.g., style [81, 82], pose [83], and identity [84, 85]) as well as diverse input modalities, such as
108 image-to-image generation. A variety of Latent Diffusion Models built on top of SD—including
109 Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89]—have also emerged, spanning a
110 broad design space of architectures and training pipelines.

111 2.3 Datasets for Generative Image Models

112 The recent success of generative vision models across tasks and domains is largely attributed to
113 the abundance of computational resources and large-scale internet datasets [90–92]. Specifically,
114 LAION-5B [93] has played a central role in the training of a broad range of open-source text-to-image
115 models, including SD and its many derivatives. This dataset, scraped from Common Crawl [94],
116 contains over 5B image-caption pairs, 2.3B of which are in English. Other prominent large-scale
117 datasets include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M image-caption
118 pairs, respectively.

119 As LAION-5B gained widespread adoption, concerns grew over its biases [97–101]. Despite filtering
120 attempts, harmful content persisted [97–99], including NSFW material [100] and hundreds of CSAM
121 instances [101], prompting its temporary removal from official channels. More broadly, the dataset
122 also suffers from low-quality images [102] and internet-style captions (e.g., product descriptions)
123 that misalign with how users typically prompt trained models [103].

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks to find sparse representations of input data, where each sample can be reconstructed using a linear combination of a few dictionary atoms [104–106]. Built upon compressed sensing theory [107–108], the field has evolved from early vector quantization methods [109] into a broad family of approaches including Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. Recent advances span online methods [114], structured sparsity [115], and theoretical guarantees [116–117], alongside increasingly broad connections to deep learning [118–119].

Advances in sparse coding have also been leveraged to study the emergence of high-level concepts inside diffusion models [120]. Prior to diffusion models, concept-grounded interpretability had been applied to a range of earlier generative architectures through concept-bottleneck models, which require human intervention at training time [121], and post-hoc detectors that retrofit concept supervision [122]. However, both of these approaches require human-defined concepts and hence inherently miss the broader trends that a user does not explicitly register.

Aggregate metrics (e.g., precision, recall, density, and coverage [35, 123]) and latent density scores, which predict sample quality based on the model’s latent space [124], have emerged as general tools for evaluating generative image model capabilities. While these effectively uncover distributional gaps, they offer little insight into the specific concepts that are under- or over-represented.

Together, these limitations motivate a scalable, unsupervised, and broadly applicable framework that can systematically identify and quantify concept-level failure modes across generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We begin by formalizing the general notion of *conceptual blindspots*: systematic discrepancies between the conceptual content of natural images and that of model-generated outputs. This formulation enables us to derive principled, quantitative measures that characterize which concepts are under- or over-represented by a generative model relative to its data distribution, in a manner that applies to any such model. The overall process is illustrated in Fig. 2. While we rely on standard assumptions in this pursuit [126–130], we empirically find that meaningful phenomenology is elicited across models even when these assumptions are relaxed or violated, suggesting the framework’s applicability extends well beyond its idealized conditions.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G: \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational simplicity, we use $p(\cdot)$ to denote both the latent density $p(c)$ and its push-forward to image space $p(x)$, where $x = G(c)$. This is justified by the invertibility of G , which induces a valid distribution over $\mathcal{X}$ via the change of variables formula. In essence, the individual dimensions of the latent space reflect the broad set of **Concepts** underlying the data-distribution $P_{\mathcal{X}}$, defined over some observation space of images $\mathcal{X}$. For example, different latents may correspond to a wide variety of concepts such as **color**, **shape**, **size**, **location**, and so on [131–132]. We also let the data-generating process associate a text-description $t \in \mathcal{T}$ with any image sampled from the data distribution, but do not explicitly model it. These text descriptions can then be used to train any text-conditioned **Generative image model** g_θ , with parameters θ , on a set of image-text pairs, mapping a noise signal $\eta \sim \mathcal{N}(0, I)$ and a text-description of the scene t to produce images x' illustrating the latter.

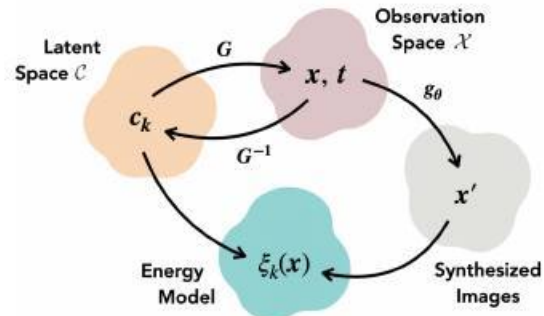


Figure 2: **Intuitive mapping of our general framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) through the (invertible) DGP. S_θ denotes an arbitrary generative model that produces images x' based on t . The energy model $\xi_k(x)$ extracts concept representations from both x and x' , enabling comparison of concept distributions to identify blindspots for any such model.

To define conceptual blindspots in any model g_θ , we must assess the probability mass assigned to a concept by the data-generating process and compare it to the mass assigned by the model. To this end, we take an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$ and define a corresponding set of synthetically generated images $\mathcal{D}'_\mathcal{X}$ using the text descriptions. To estimate the probability of occurrence of a concept, we consider an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps images to a d -dimensional vector, where each dimension associates a scalar representing the energy in the k^{th} concept, i.e., $\xi_k(\mathbf{x}) = E(c_k)$. These estimates are feasible because we assume the data-generating process is invertible. Correspondingly, the probability the data-generating process associates with the image $\mathbf{x}$ can then be defined as $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (where Z_ξ is the partition function), hence yielding a population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{\mathbf{x} \in \mathcal{D}_\mathcal{X}} p_k(\mathbf{x})$. Using this and the sigmoid map $\sigma(\cdot)$, we define below the average energy difference in the k^{th} concept between the datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$.

Definition 2 (Energy Difference). Let $\mathbf{x} \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-generating process G , and let $\mathbf{x}' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

Thus, the energy difference in the k^{th} concept describes, for any concept, the ratio of the probability that it occurs in a set of observations (here, $\mathcal{D}'_\mathcal{X}$) relative to a baseline dataset (here, $\mathcal{D}_\mathcal{X}$). Based on this general measure, we can now define conceptual blindspots as follows.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

Overall, we define a conceptual blindspot, generally, as any concept whose likelihood of occurrence in generated images deviates markedly—either through suppression or exaggeration—from its prevalence under the data-generating process. Suppressed concepts exhibit disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). Throughout our analysis, we adopt threshold values of $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate these regimes, though the definition itself is agnostic to the particular choice of thresholds.

We also note that this definition generalizes the idea of “mode collapse” studied in past work (e.g., see Bau et al. [39]): the key difference lies in the granularity at which the analysis is performed. Specifically, mode collapse focuses on exaggerated / suppressed odds of generating *entire images*, whereas our formulation captures the more general case of changed odds for arbitrary specific concepts. For example, if a model fails to produce images of an object with a `white` background, we say this concept is a suppressed conceptual blindspot.

4 Method: Operationalizing the Definition of Conceptual Blindspots

We next describe our pipeline for identifying conceptual blindspots in an arbitrary generative model g_θ . As per Sec. 3, the salient objects we need for this are (i) a set of images sampled from g_θ that allow comparison with the ground-truth generative process, and (ii) an energy model that enables said comparison; both components are model-agnostic and apply uniformly regardless of the underlying generator. Below, we use $\|\cdot\|_F$ to denote the Frobenius norm and $\|\cdot\|_0$ to denote the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ implies element-wise non-negativity. For $n > 0$, we let $[n] := \{1, \dots, n\}$, and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

From Prompts to Latent Representations. To identify conceptual blindspots in any model g_θ , we compare a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs $(\mathbf{x}, \mathbf{t})$ sampled from the data-generating process

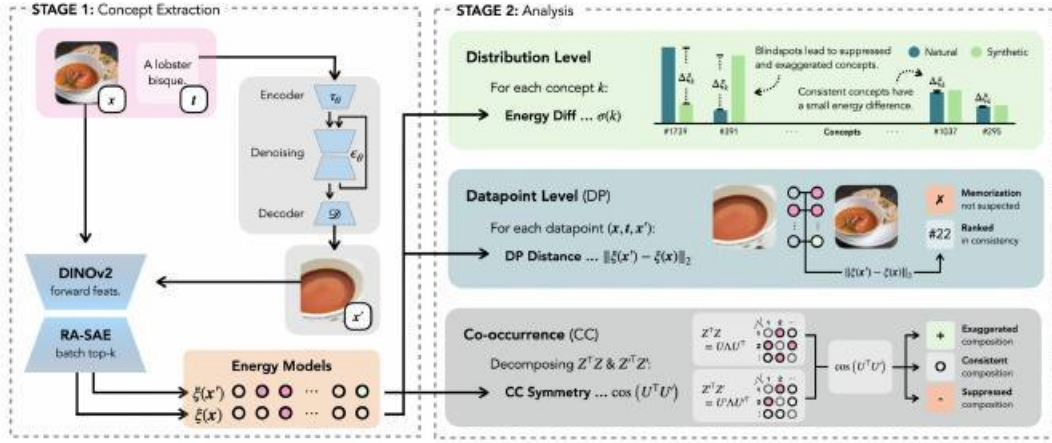


Figure 3: **Concept Extraction Pipeline.** For any triplet (x, t, x') , the concepts in x and t are extracted by obtaining each image’s DINOv2 features, which are further processed by a RA-SAE into sparse concept embeddings, yielding energy models $\xi(x)$ and $\xi(x')$, respectively. In particular, $\xi_k(x) = E(c_k)$ holds the energy in the k^{th} concept. The same pipeline applies uniformly across generators and concept types, against their synthetic counterparts sampled from the generative model g_θ using the same text descriptions.

Specifically, given t , we synthesize a counterpart image x' using a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$; in our experiments this is instantiated as a denoising diffusion probabilistic model (DDPM) [78, 86, 87], though the pipeline places no restriction on the generator’s specific form. Sampling occurs in latent space via a reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, I), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, I),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0 is decoded via a pretrained VAE to yield the synthetic image $x' = \text{VAE}(\gamma_0)$. Crucially, for the remainder of the paper we treat g_θ purely as a black box that maps prompts to images, $t \mapsto x'$, so that the approach applies to any generator irrespective of its internal architecture or sampling procedure.

Defining the Energy Model. Building on a broad line of prior work showing that self-supervised learning methods can invert the data-generating process and recover the underlying energy function up to linear transformations [126, 128, 133, 134], we adopt DINOv2 [135]—a state-of-the-art self-supervised model—as a general-purpose feature extractor for our analysis. Under the expectation that the number of concepts underlying the DGP is far larger than the dimensionality of the model’s feature space [51, 136], we train sparse autoencoders (SAEs) on its features to isolate subspaces corresponding to these concepts [50, 56, 137–139]. The intuition here is that if the concepts underlying the generative process are modeled via approximately orthogonal directions by DINOv2 (as assumed in our independence constraint in Def. 1), then an SAE should be able to isolate a broad range of these concepts along individual dimensions in its latent space [136]. The activation assigned by the SAE to a given dimension then serves as our approximation of the ground-truth energy function for the concept modeled by that dimension.

Formally, using $f : \mathcal{X} \rightarrow \mathbb{R}^d$ to denote our feature extraction module (i.e., the DINOv2 model), we extract features $a = f(x) \in \mathbb{R}^d$ from both natural and synthetic images drawn from the datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$, regardless of which generator produced the latter. Assuming the count of image-text pairs is n , we stack the real and generated features into matrices $A, A' \in \mathbb{R}^{n \times d}$. We then decompose each feature vector into a sparse combination of learned concept atoms using an SAE. Specifically, let $D \in \mathbb{R}^{d \times K'}$ denote a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the SAE encoder that maps input features to sparse codes. Applying Ψ row-wise yields the matrix of activations $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, where each row $z_i = \Psi(a_i)$ represents the concept decomposition of an image. The decoder reconstructs features via $ZD^\top$, and the SAE is trained to minimize the reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, D} \|A - \Psi(A)D^\top\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \quad \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs often drift toward arbitrarily oriented dictionaries, making downstream analyses highly sensitive to the random seed. To mitigate this instability and make our study reproducible and independent of the seed—a property essential for comparisons that generalize across models—we employ the *Archetypal* SAE (RA-SAE) [56] under a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary $\mathbf{D}$ to be a convex combination of training data. Specifically, we write $\mathbf{D} = \mathbf{W}\mathbf{A}$ with $\mathbf{W} \in \Omega_{K',n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K',n} := \left\{ \mathbf{W} \in \mathbb{R}^{K' \times n} \mid \mathbf{W} \geq 0, \mathbf{W}\mathbf{1} = \mathbf{1} \right\}. \quad (3)$$

Thus every atom $\mathbf{D}_i$ lies in the convex hull of the data $\text{conv}(\mathbf{A})$, and any reconstruction $\mathbf{Z}\mathbf{D}^\top$ resides inside the conic hull of the data $\text{cone}(\mathbf{A})$. This ensures learned concepts remain faithful to the support of the data distribution [56]. Once trained, the SAE provides a consistent set of sparse codes for any generator: $\mathbf{Z}$ for real images and $\mathbf{Z}'$ for their generated counterparts. *These codes capture the same prompt-conditioned visual semantics in terms of shared, interpretable concepts, with the activation value of each concept serving as the energy value for our analysis of conceptual blindspots.* In summary, our method defines a general, structured pipeline that, given a prompt and its associated real image (t, x) , produces two sparse concept vectors (z, z') , enabling direct comparison of the real and generated visual content in a common conceptual basis—independently of which model generated the synthetic image.

This summarizes our full pipeline: starting from a captioned image (t, x) , we synthesize a counterpart x' and map both images into a shared, sparse concept space via a vision encoder and a trained SAE, yielding interpretable representations (z, z') that serve as a common foundation for evaluating the conceptual shifts induced by any generative process.

5 Results

As a representative instantiation of our framework, we analyze four generative image models spanning distinct architectures and training pipelines, all trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). Our analysis spans three complementary levels (Fig. 3): a **distribution-level** evaluation reveals suppressed and exaggerated concepts; a **datapoint-level** analysis surfaces failures tied to ambiguity, omission, and memorization; and a **compositional** analysis uncovers subtle distortions in concept co-occurrence geometry. Together, these levels illustrate the breadth of phenomena the framework can surface.

Our core contribution is an interactive exploratory tool, shown in Fig. 5. The tool is a general-purpose, web-based interface built around a UMAP projection of concept representations, enabling visualization and comparison of concept-level energy differences across models. It is publicly available at a software repository, along with pre-computed energy difference data for the four models evaluated in this work (SD 1.5, SD 2.1, PixArt, and Kandinsky), and is readily extensible to additional models. All subsequent analyses in this paper are derived from insights enabled by this tool. Its primary functionalities, which support these analyses, include:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization spanning the full set of 32,000 concepts from the RA-SAE, and supports side-by-side comparison across arbitrary model families. Each scatter point represents an individual concept, color-coded by its energy difference.
- **Inspect concepts.** Every concept has a card with key statistics, representative real and generated images (x, x') , and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond the UMAP and per-concept views, the tool features global rankings of suppressed and exaggerated blindspots, surfacing the most notable conceptual blindspots for any evaluated model.



Figure 4: **Representative image-caption pairs** from LAION-5B, with matching images generated from the same prompt by a diverse set of models—SD 1.5, SD 2.1, PixArt, and Kandinsky.

309 Rather than exhaustively studying any single phenomenon, we present a set of high-level findings
 310 chosen to highlight the tool’s versatility and to enable broad, customizable exploration across models
 311 and concepts.

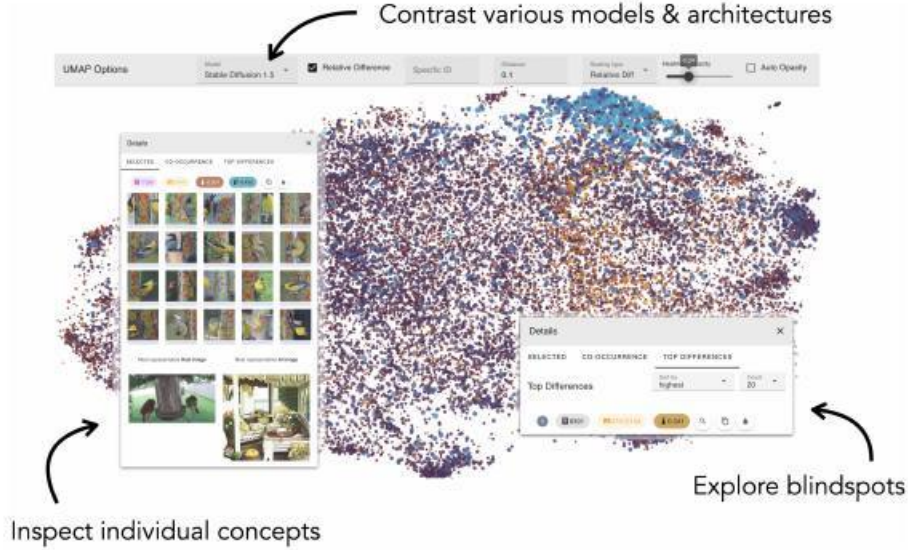


Figure 5: **Overview of the Exploratory Tool.** The general-purpose web interface displays a UMAP projection for each evaluated model, where each dot represents a concept, color-coded by its energy difference. When a concept is selected, a detail panel presents illustrative images, statistics, and the most representative natural and generated images x and x' . An ordered list of the concept’s co-occurrences is shown alongside global rankings of blindspots. The same interface generalizes to any model whose generations are processed through our pipeline.

312 5.1 Generative Image Models Cannot Produce Unseen Concepts

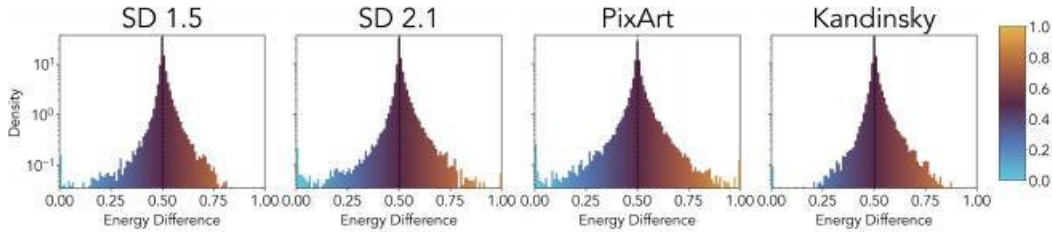


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ across the full set of 32,000 concepts, comparing the natural and synthesized distributions for each of the diverse evaluated models. Values left of zero represent suppressed concepts (under-represented); values right of zero represent exaggerated concepts (over-represented).

313 To assess disparities between the generative models and the natural image distribution, we begin by
 314 analyzing the marginal energy difference $\delta(k)$ across the full set of 32,000 concepts learned using
 315 RA-SAE. As defined in Sec. 4, this quantity reflects the relative prevalence of each concept in the
 316 synthesized versus natural image sets. A value of $\delta(k) < 0.1$ indicates that concept k is under-
 317 represented (suppressed) in the generated images, while $\delta(k) > 0.9$ indicates over-representation
 318 (exaggerated). Fig. 6 presents the distribution of $\delta(k)$ for each of the four evaluated models spanning
 319 distinct architectures. Across all of these models, we observe heavy-tailed histograms with substantial
 320 mass at both extremes, pointing to systematic, rather than incidental, discrepancies in concept
 321 coverage. Notably, the left tail—corresponding to suppressed concepts—is consistently denser and
 322 longer than the right, indicating a broadly shared tendency toward concept suppression. This
 323 asymmetry is reflected in the negative skewness observed across all distributions: Skewness = -0.54
 324 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

We also note that while all models—despite their architectural differences—exhibit both suppressed and exaggerated concepts, their specific profiles vary. For instance, PixArt shows a wider spread, suggesting a more heavily suppressed concept distribution. Nevertheless, the consistent left-skew observed uniformly across models underscores a common, cross-architecture tendency toward concept omission; the finer characteristics of this behavior require further analysis, which we pursue in the next Sections.

5.2 Structure and Specificity of Conceptual Blindspots

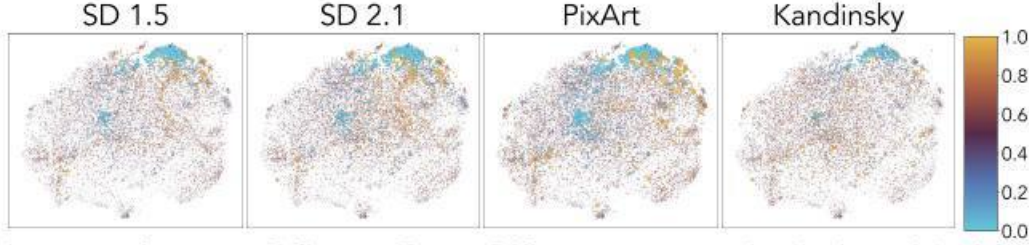


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of the full set of 32,000 concepts, colored according to their energy difference $\delta(k)$ between the natural and synthesized distributions, shown for each of the diverse evaluated models. Clusters reveal patterns of conceptual blindspots, with suppressed concepts on the blue end and exaggerated ones on the yellow end.

While the previous section quantified marginal discrepancies in concept frequency, here we investigate their global structure by embedding the full set of 32,000 concepts into two dimensions using UMAP on the sparse codes, coloring the concepts by their $\delta(\cdot)$ values. As shown in Fig. 7, distinct clusters of concepts emerge consistently across all models. These clusters often correspond to contiguous regions of conceptual blindspots, especially for suppressed (blue) concepts, suggesting that blindspots are broadly structured rather than random—reflecting shared biases in either training distributions or architectural priors. To assess how consistently these blindspot patterns hold across models, we further analyze both the magnitude and structure of concept-level $\delta(\cdot)$ values. Fig. 8 presents scatter plots and pairwise Pearson correlation coefficients between the $\delta(k)$ vectors of SD 1.5 and all other models. As expected, SD 1.5 and 2.1 exhibit strong correlation ($r = 0.82$), reflecting their shared architectural and training pipelines. In contrast, their correlations with the more architecturally distinct PixArt and Kandinsky are substantially lower— $r = 0.41$ and $r = 0.46$, respectively—indicating that models across the broader design space emphasize different regions of the conceptual space.

Overall, the analysis above reveals that while some blindspots are universally shared across the diverse models we study—likely due to properties of the dataset—others are highly model-specific, emerging from idiosyncrasies in training dynamics or model capacity. This dual structure motivates the need to identify and study, in subsequent sections, both blindspots that are broadly shared across models and those that are unique to specific models.

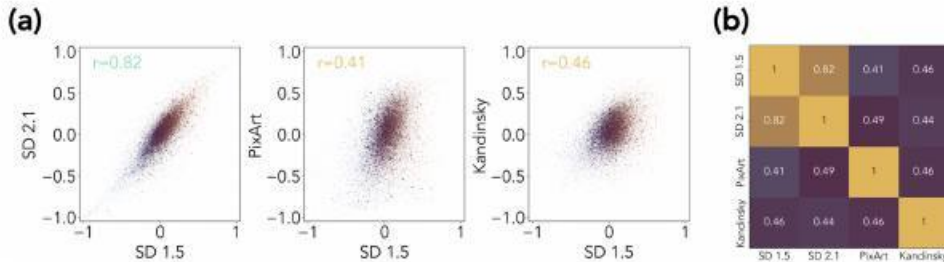


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported top left. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations among the more distinct architectures, revealing both shared and model-specific blindspots. **Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlation coefficients for δ between all models, quantifying the degree of shared conceptual blindspots across this diverse set of models.

5.3 Qualitative Examples of Blindspots

We next visualize specific examples of both suppressed and exaggerated blindspots to gauge the range of concepts that fall under these regimes. Specifically, in Fig. 9a we show an example of a conceptual blindspot suppressed universally—we find that all evaluated models, despite their architectural diversity, fail to reproduce the concept `solid white on documents`. As can be seen in the figures, despite the caption explicitly referencing this concept, none of the generated images reflect the intended visual semantics, suggesting that this region of the concept space is systematically under-sampled across models. Meanwhile, Fig. 9b highlights a model-specific blindspot: the concept `pan` is accurately captured by three of the models, yet conspicuously missing from generations produced by Kandinsky. This reinforces the findings from Sec. 5.2 where cross-model agreement was found to be high in some cases but limited in others.

Conversely, in Fig. 10 we present a case of exaggeration, where the concept `shadow under animal` is overly emphasized in generated images. While shadows are mildly plausible, their consistent and pronounced rendering across the diverse models—relative to the more nuanced and variable occurrences in natural images—suggests a broadly overactive prior. Interestingly, despite systematic attempts to find concepts uniquely exaggerated by any single model, we did not identify any clear examples—suggesting that exaggerations tend to be approximately universal across models.

Overall, the examples above concretely demonstrate how conceptual blindspots manifest in generated outputs, illustrating that our energy-based diagnostic can surface both broadly shared and model-specific failure modes within a single, general framework. Notably, it enables the identification of surprising model limitations—such as the consistent failure to reproduce clear or solid background elements, like whitespaces on documents, across all evaluated models. This raises the broader possibility that certain failure patterns stem from architectural constraints or training-data biases that transcend the idiosyncrasies of any individual model.

While these aggregate-level analyses are informative, they invite a deeper and more general question: do these blindspots emerge only in the aggregate across many samples, or do they manifest even at the level of individual datapoints? This finer-grained perspective allows us to probe the mechanisms behind blindspots more



Figure 10: **Example of an Exaggerated Conceptual Blindspot.** Four synthesized images $\mathcal{D}'_{\mathcal{X}}$, archetypal for the concept `shadow under animal`, alongside the corresponding natural image $\mathcal{D}_{\mathcal{X}}$ and caption t used to generate it. Note the accentuated, nearly solid black shadows underneath the animals. We deem this blindspot universal since it is present across all evaluated models, spanning distinct architectures.

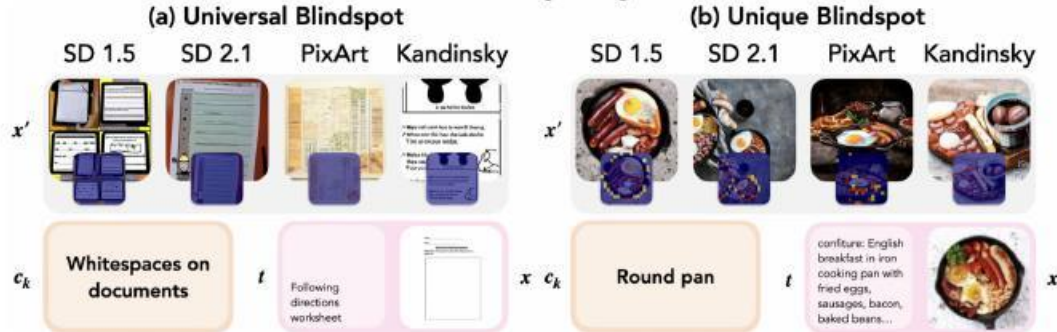


Figure 9: **Examples of Suppressed Conceptual Blindspots.** The natural images x , representative c_k and t , shown alongside four synthesized images x' , generated using S_{θ} . The universal blindspot is present across all evaluated models regardless of architecture; the unique blindspot appears only in Kandinsky.

389 directly—uncovering cases of prompt mis-
 390 interpretation, latent memorization, or
 391 both.

393 5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

394 To move beyond population-level statistics, we apply the same framework at the
 395 level of individual natural vs. generated image pairs, examining those for which the
 396 $\delta(\cdot)$ values averaged across all concepts exhibit the largest and smallest differences.
 397 This analysis facilitates a broad understanding of model successes and failures, the latter of
 398 which we find often arise from prompt ambiguity or memorization artifacts. For example,
 399 Fig. 11a shows instances with near-zero difference in average $\delta(\cdot)$ values. In these cases, the
 400 generated images are conceptually indistinguishable from the original. However, qualitative
 401 inspection clearly shows that this occurs not because the model faithfully captures the prompt
 402 semantics, but through pure replication of memorized templates: we observe repetitive visual
 403 structures (e.g., outlines of clothing or object arrangements), indicating that the model may
 404 be copying from overly frequent patterns in the training data. By contrast, Fig. 11b illustrates
 405 samples among the largest $\delta(\cdot)$ values. These indicate significant conceptual divergence be-
 406 tween the synthesized and natural image. While some of these discrepancies can be attributed to
 407 underspecified or noisy captions, others reveal genuine blindspots: the prompt describes a clear
 408 concept faithfully present in $\mathcal{D}_X$, yet the model fails to realize it in $\mathcal{D}'_X$. This failure suggests that, even when language grounding is adequate, certain
 409 concepts fall broadly outside the model’s generative abilities.



Figure 11: **Datapoint-level Conceptual Alignment.** (a) Examples with minimal energy differences, where models appear to memorize training patterns. (b) Examples with large differences, where significant concept divergences arise from prompt ambiguity or model limitations. Both regimes are surfaced by the same general datapoint-level diagnostic.

5.5 Analyzing Post-Training Effects

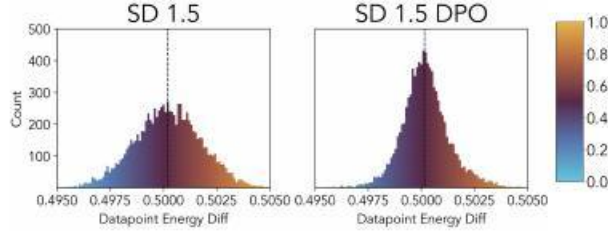


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distributions of SD 1.5 models with and without DPO, illustrating how the framework can characterize a broad class of post-training interventions.

Post-training protocols, e.g., safety fine-tuning, have broadly been argued to reduce the diversity of model generations [140]. Given our pipeline’s general ability to isolate differences between a model’s generations and the ground-truth DGP, we next apply it to understand the effects of DPO—a widely used safety fine-tuning protocol [141]—as a representative post-training intervention. Specifically, we compare two variants of the SD 1.5 model: one trained with DPO, and one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of the difference between their internal concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents a histogram of these datapoint-wise energy differences. The DPO-enhanced model exhibits both a lower median and a narrower spread, indicating a more consistent match between the generated concept distribution and the ground-truth DGP. This suggests that DPO may serve to regularize the model’s concept distribution, encouraging outputs that more closely reflect the semantic content of the seen inputs. While our analysis does not disentangle the specific inductive biases introduced by DPO, these results provide empirical evidence that its optimization objective, which favors human-preferred generations, indirectly promotes a better match with the training distribution. In particular, it reduces *both over- and under-* activation of individual concepts relative to the natural baseline. More broadly, these findings highlight the utility of our pipeline for characterizing the downstream effects of a wide range of post-training interventions: not merely in terms of output quality, but in how they reshape the conceptual geometry of the model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

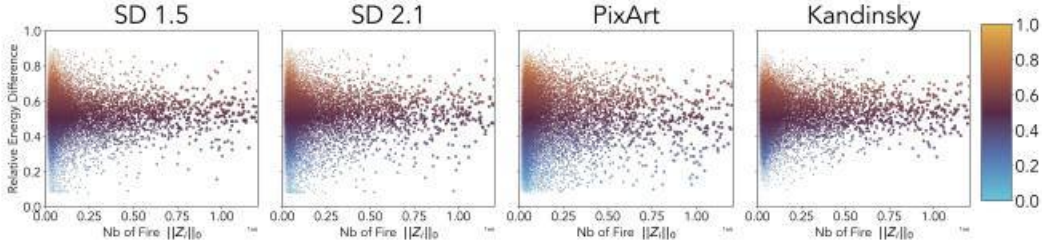


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots showing the relationship between concept frequency (x-axis) and the energy difference (y-axis) across the four diverse evaluated models. Each point represents a concept, with size proportional to its activation frequency.

We previously hypothesized that conceptual blindspots are not merely architectural artifacts, but may also emerge as a direct and general consequence of distributional peculiarities of certain features. In this section, we empirically test this hypothesis by examining whether concepts that are rarely activated in natural images, i.e., those in the long tail of the data distribution, tend to give rise to blindspots across generative models. Specifically, we process the natural dataset $\mathcal{D}_X$ through the trained SAE and compute, for each concept k , its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$, where $\mathbf{Z}_{:,i}$ denotes the activations of concept i across all our images. We then correlate this with the absolute energy difference observed across generated outputs. Fig. 13 visualizes this relationship for all evaluated models. Consistently across models, we find that concepts with higher frequency in natural data tend to show lower energy discrepancies, while rare concepts—especially suppressed ones

452 ($\delta(k) < 0.5$)—exhibit significant alignment errors. This suggests that many blindspots stem not from
453 randomness or isolated model quirks, but from systematic effects broadly tied to long-tail concept
454 distributions. Addressing these issues may therefore require not only architectural changes but also
455 more general strategies such as data reweighting or augmentation.

456 6 Discussion

457 Our analysis reveals multiple conceptual blindspots across four popular generative image models
458 spanning distinct architectures. The results presented here, however, only scratch the surface: each
459 individual finding could warrant its own dedicated investigation. Rather than delving deeply into any
460 one of these questions, we instead showcase the broad versatility of our method and exploratory tool.
461 Out of the box, and without task-specific tuning, they enable systematic identification of concepts
462 that models struggle to generate, detection of memorization artifacts, discovery of datapoints with
463 insufficient captions, quantification of post-training effects, and characterization of conceptual shifts
464 across model architectures. We thus open space for follow-up work to extend the depth of analysis,
465 broaden the range of evaluated architectures and modalities, and probe the root causes of conceptual
466 blindspots. Since the pipeline treats the generator as a black box, it applies readily to model families
467 beyond those studied here. Future work could also explore hierarchical representations of concepts in
468 RA-SAE to enable an even more nuanced analysis.

469 **Limitations.** We wish to highlight several limitations of our work, which also delineate the
470 boundaries within which the framework generalizes. By relying on DINOv2 and RA-SAE for
471 concept extraction and representation, our approach is inherently constrained to the broad—but not
472 exhaustive—range of concepts these models capture; concepts poorly represented by them will escape
473 our analysis, though the pipeline itself is agnostic to the specific choice of encoder and could be
474 paired with alternatives. Additionally, while our sample size of 10,000 images is substantial and
475 sufficient for reliable estimation (see Appendix), it may not fully capture the long tail of rare concepts,
476 concept co-occurrence, or other compositional statistics.

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Appendix

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A Experimental Setup

This section details the experimental setup for our analysis of four popular generative image models spanning distinct architectures: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. These models serve as a representative instantiation, and the same setup extends directly to other generators. The code is available at a software repository.

A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our broad domain of natural images. Due to concerns with CSAM and other unsafe content in the dataset, the original data release is no longer available. A substitute release of a subset of this dataset with additional filtering of the unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe> is used.

The sampling procedure consists of: (1) loading the full LAION dataset using the Hugging Face datasets library, (2) performing validation to ensure proper URL structure and resource availability via HTTP HEAD requests, and (3) employing random sampling with replacement until reaching the target count of 10,000 valid samples. This general-purpose approach, which yields D_G with (x, t) tuples, ensures our observation space contains accessible image-text pairs for comparative analysis, even though a large portion of the original image URLs has become unavailable since release. Additional examples of synthesized images are shown in Appendix H.

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , yielding triplets (x, x', t) . Specifically, given the 10,000 image-text pairs (x, t) from D_G , we use t to synthesize counterpart images x' with each generative model g_θ .

The synthesis process follows the standard text-to-image generation pipeline appropriate to each model architecture, implemented using the Hugging Face diffusers library, with models loaded at mixed precision (fp16). To keep the comparison uniform across this diverse set of models, all synthetic images are generated at 512×512 pixel resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/ruwnayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>.

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1. We compute energy differences $\delta(\cdot)$ across all 32,000 concepts for each evaluated model. The sigmoid transformation with temperature $T = 0.8$ is applied during normalization. The resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 We embed the complete set of 32,000 concepts into two-dimensional space using UMAP applied to the sparse concept codes. Each point in this UMAP represents an individual concept, colored according to its energy difference $\delta(\cdot)$, emphasizing both suppressed and exaggerated blindspots. To quantify how consistently blindspots hold across models, we compute pairwise Pearson correlation coefficients between the $\delta(\cdot)$ vectors of all model pairs, producing both scatter plots and correlation matrices. This analysis reveals whether blindspots cluster in conceptual space and distinguishes model-specific from universal patterns of conceptual blindspots.

Section 5.3 We rank the full set of 32,000 concepts by their energy difference $\delta(\cdot)$, and manually examine the extrema (both suppressed and exaggerated blindspots). For suppressed blindspots, we select concepts with $\delta(\cdot) < 0.1$; for exaggerated blindspots, we choose those with $\delta(\cdot) < 0.9$. The presented examples are manually annotated with textual descriptions of the respective concepts through inspection of their most activating images and spatial attention patterns. We outline ongoing efforts to automate this concept interpretation, so that it can scale to the full concept space, in Appendix D

Section 5.5 As a representative post-training comparison, we contrast SD 1.5 with and without DPO in the following fashion: for each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. We apply a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before taking their mean. This yields datapoint-wise energy differences that quantify how much each generated image deviates from its natural counterpart in concept space. Finally, these differences are visualized as overlapping histograms, contrasting both model variants.

Section 5.6 For each concept c_k , its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$ (the count of non-zero activations across the natural dataset) is counted. A sigmoid normalization with temperature $T = 0.4$ is then applied to the energy differences $\delta(\cdot)$. The analysis is visualized using scatter plots where the x-axis is the empirical concept frequency and the y-axis is the sigmoid-transformed energy difference. The point sizes are proportional to activation frequency and point colors are proportional to the magnitude of energy differences.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. This yields a general, scalar measure of conceptual divergence for each image pair. The samples are then ranked by their energy differences. Minimal divergence indicates potential memorization artifacts, while maximal divergence points to significant conceptual failures. This analysis enables qualitative inspection of a broad range of specific failure modes.

A.5 • Co-occurrence Analysis

Appendix F For both the natural and synthesized data D_G and D_{g_θ} , concept co-occurrence patterns are analyzed through the co-activation matrix $Z^T Z$, which captures pair-wise correlations in concept usage. Spectral analysis via eigendecomposition is performed to examine the dominant conceptual directions. The alignment between natural and synthetic co-occurrence structures is assessed using cosine similarity heatmaps between the top-100 eigenvectors of each co-occurrence matrix. These 100×100 similarity matrices are visualized as square heatmaps, where perfect diagonal alignment would indicate identical principal concept axes, while off-diagonal patterns would reveal rotations and mismatches in compositional geometry.

B Custom RA-SAE Training Details

This section summarizes the training details of our custom archetypal SAE (RA-SAE) [56]. The model is open-sourced at a software repository, and being trained on a broad, general-purpose corpus is directly reusable across the diverse generators we study.

Dataset. The auto-encoder is trained on the complete ImageNet-1k training split (≈ 1.28 M RGB images), providing broad coverage of natural-image concepts. Each image is converted to 261 visual tokens using DINOv2 [135]; tokens are fed to the SAE without class or position embeddings. The total number of training tokens is therefore $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

Dictionary. The dictionary has 32,000 concept dimensions. For the sparse activation rule, top- k masking with $k = 5$, is used; activations outside the largest five per input are set to 0. The weights are initialized using Xavier/Glorot. The training is conducted at mixed precision (fp16), with the last ten epochs performed at full precision.

Optimizer and Schedule. The model is trained for 50 epochs using base AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) optimizer is employed with weight decay set to 10^{-5} . Linear warm-up is applied on the first 5% of steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. MSE loss is used alongside an auxiliary term penalizing activations that never enter the top- k set, where $\lambda = 10^{-5}$.

1010 C Computational Resources

1011 This section summarizes the GPU resources used for training and experiments in support of this
1012 paper, indicating the modest and broadly attainable compute footprint of the overall approach. In
1013 total, we used approximately 202 GPU-hours on NVIDIA H100s and H200s.

1014 **RA-SAE.** Trained for approximately 24 GPU-hours on three NVIDIA H100s.

1015 **Synthesized Images.** Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator
1016 when distributed across four NVIDIA H200 GPUs. With five generators, this totaled approximately
1017 100 GPU-hours on a NVIDIA H200.

1018 **● Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3
1019 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

1020 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
1021 datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one
1022 H200 GPU (total ~ 15 GPU-hours).

1023 D Autointerpretability

1024 In an effort to automate the interpretation of concepts identified by the RA-SAE—a prerequisite for
 1025 scaling our analysis to the full concept space and to arbitrary models—we qualitatively evaluated the
 1026 ability of Vision-Language Models (VLMs) to describe them. In particular, we queried ChatGPT
 1027 4o via the OpenAI API using a variety of prompts and formats of an image x'' in which the given
 1028 concept was salient, recording the description generated by the VLM. In this section, we highlight
 1029 the findings from our exploration. For evaluation purposes, we allowed the VLM to provide its full
 1030 reasoning; to turn this into a scalable solution, a simple modification of the prompt—asking the VLM
 1031 to provide only the description—would filter out the reasoning.

1032 First, we provided the VLM with two images: a raw version of x'' and x'' with a heatmap superim-
 1033 posed to localize the intensity of the concept:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DiNOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DiNOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

1034 As seen in the example, the VLM did not interpret the heatmap visualization correctly. Instead of
 1035 describing the wood texture in the background—the actual salient region—it described the ring in the
 1036 foreground, which received no attention from the RA-SAE.

1037 To determine whether this was merely a prompting issue, we tested the same scenario again—this
 1038 time using an expanded prompt that explained the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DiNOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DiNOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

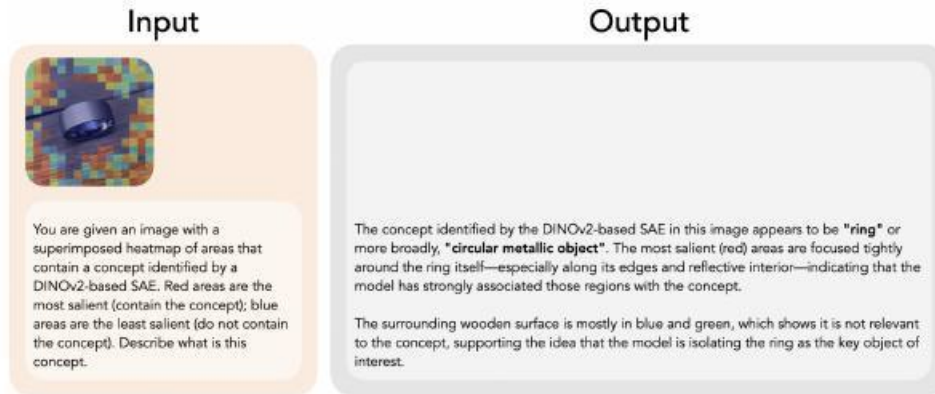
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

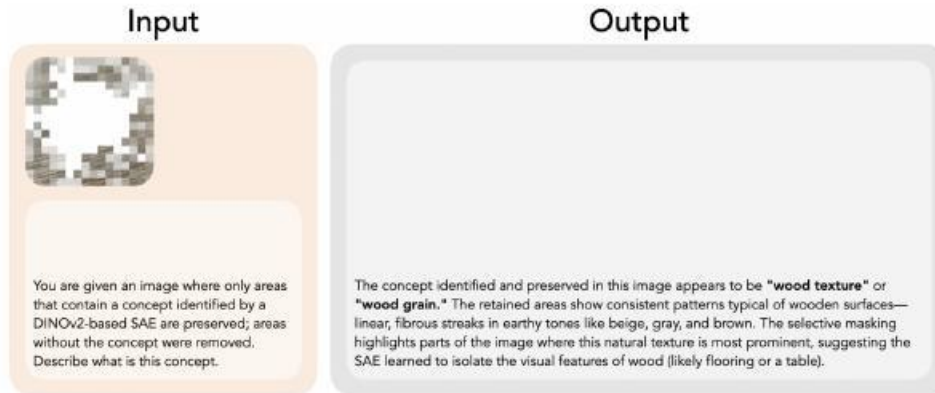
1039 This prompt modification did not resolve the issue, as the VLM still identified the ring as the
 1040 prominent concept in the image. A range of other prompting techniques, including those describing
 1041 the heatmap in even greater depth (e.g., by including the code used for its generation), were explored
 1042 with similar results.

1043 We next tested providing the VLM only with the heatmap-superimposed x'' and a similar prompt
1044 explaining how to interpret the heatmap:



1045 Once again, the result failed to identify the wooden background as the prominent concept in the image
1046 and instead focused on the ring. As in the previous case, we experimented with various prompting
1047 strategies using this image format, but the results remained largely unchanged.

1048 From these observations, we concluded that, in general, the heatmap alone is insufficient to guide
1049 the VLM's attention to the salient areas. The presence of less relevant regions appears to distract the
1050 model. To address this, we replaced the heatmap-superimposed x'' with an alpha-masked version of
1051 x'' :



1052 As shown in the example, the VLM was now able to correctly identify wood texture as the dominant
1053 concept. We observed consistent success across a broad range of concepts and images.

1054 The formatting of x'' and the prompt shown above yielded the most reliable results in our qualitative
1055 evaluation. However, we note that this evaluation is limited by its qualitative nature (due to the
1056 absence of ground truth annotations) and its focus on a single VLM; broader validation across
1057 additional models and concepts remains open. We hope future work on the autointerpretability of
1058 SAE concepts can build on and generalize this analysis.

E Additional Results: Model-Specific Blindspots

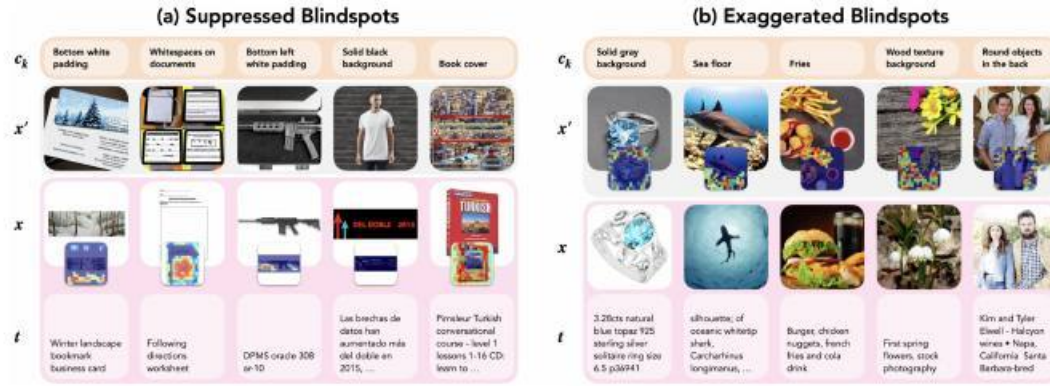


Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots) image, based on the highest absolute activation, is shown; the same analysis is applied uniformly across all evaluated models. The spatial heatmap for the concept is superimposed atop the image.

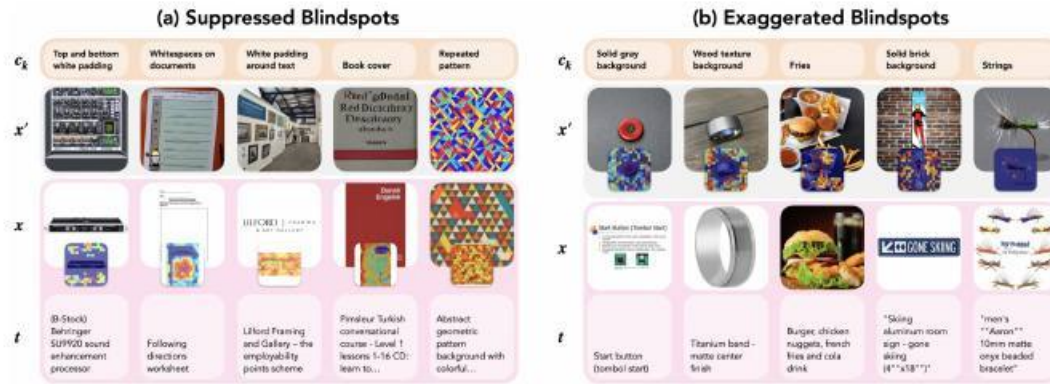


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots) image, based on the highest absolute activation, is shown; the same analysis is applied uniformly across all evaluated models. The spatial heatmap for the concept is superimposed atop the image.

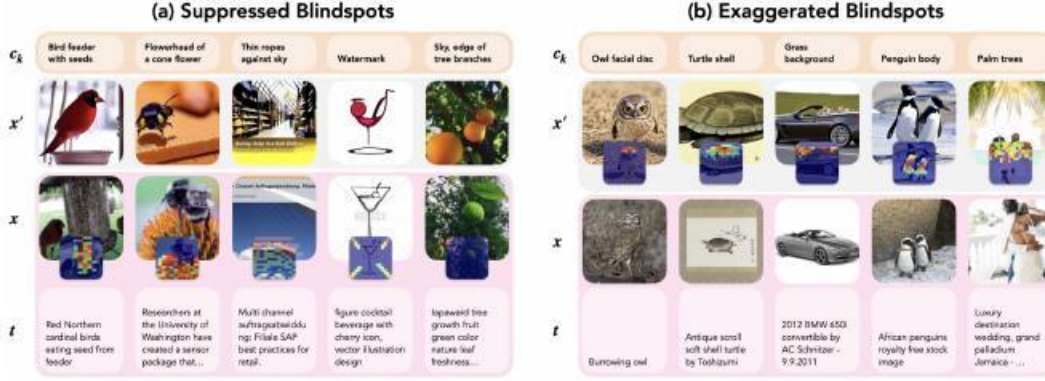


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots) image, based on the highest absolute activation, is shown; the same analysis is applied uniformly across all evaluated models. The spatial heatmap for the concept is superimposed atop the image.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots) image, based on the highest absolute activation, is shown; the same analysis is applied uniformly across all evaluated models. The spatial heatmap for the concept is superimposed atop the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, our analysis has centered on individual concept activations. Yet visual scenes are rarely composed of isolated concepts; instead, they are structured through rich co-occurrence patterns that encode compositional semantics. We now examine whether generative models, across architectures, capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which reflects pairwise correlations in concept usage.

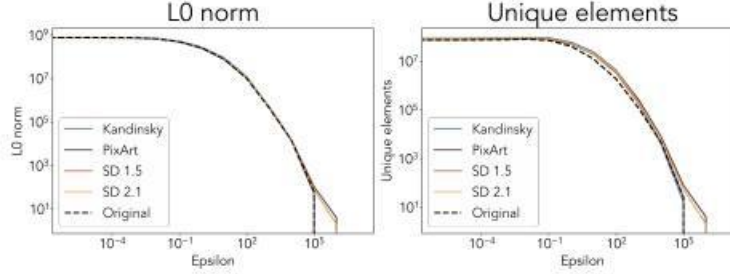


Figure 18: **Sparsity and Structural Divergence.** On the left: L_0 norm of the co-occurrence matrix $Z Z^T$ as a function of ϵ (threshold), indicating how many entries remain active in each model. On the right: number of unique entries in the synthesized distribution relative to the natural distribution. Across the diverse evaluated models, global sparsity structure is preserved, yet activation content diverges.

Surprisingly, when assessed at the level of binary structure, diffusion models broadly approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As shown in Fig. 18 (left), the ℓ_0 norm of $Z^T Z$ – thresholded at varying ϵ values – tracks closely between the natural and synthesized distributions across all evaluated models. This indicates that the gross connectivity of the conceptual graph, i.e., which concepts tend to co-activate at all, is well preserved across architectures. Formally, one can view $Z^T Z$ as the adjacency matrix of a weighted, undirected graph over concepts, where edge weights reflect co-activation strength across the dataset.

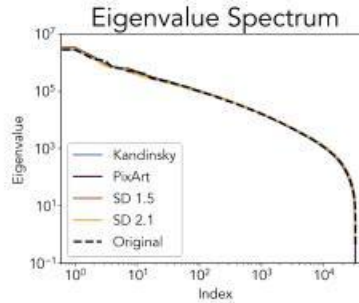


Figure 19: **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra from co-occurrence matrices $Z Z^T$ across the diverse models and the natural distribution. Across architectures, all evaluated models match the heavy-tailed decay of the natural distribution.

However, as illustrated in Fig. 18 (right), the specific content of these co-activations diverges: a substantial portion of entries in the model-generated $Z^T Z$ are not shared with the natural baseline. This suggests that while the capacity for compositionality is broadly retained, the identity of active pairings may shift, potentially reflecting model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 shows the eigenvalue spectra of the co-occurrence matrices for each model and the natural distribution. Across the board, all spectra exhibit a heavy-tailed decay consistent with power-law behavior, indicating that generative models broadly preserve the overall rank structure and variance allocation across conceptual dimensions.

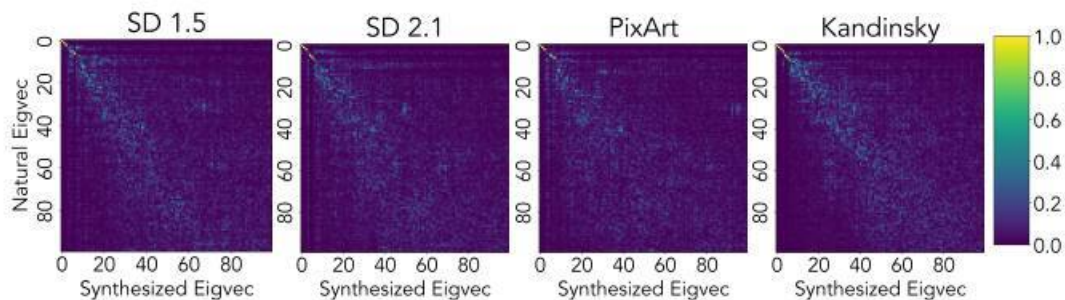


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure shows alignment of dominant conceptual directions, with varying degrees of alignment across the four diverse models.

1084 Further, we examine the alignment of dominant conceptual directions via cosine similarity heatmaps
 1085 between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20).
 1086 While all models, across architectures, exhibit partial diagonal alignment—implying overlap in
 1087 principal concept axes—the off-diagonal entries reveal rotations and mismatches in higher modes,
 1088 reflecting evident deviations in compositional geometry.

1089 Together, these findings reveal that diffusion models, across the range of architectures studied,
 1090 approximate the global shape of concept co-activation surprisingly well, yet deviate in subtle and
 1091 structured ways when examined through the spectral lens. Such higher-order discrepancies may
 1092 broadly underpin failures in generating coherent, multi-object scenes or relational concepts.

1093 G Additional Results: Blindspot Stress Testing

1094 To stress-test the blindspots identified by our method, we gathered a broad range of prompts describing
 1095 these blindspots and used them to generate many images. We then contrasted the outputs from models
 1096 in which the concept was identified as a blindspot with those in which it was not, providing an
 1097 independent, generalizable check on the method’s predictions.

1098 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1099 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1100 *text-to-image model. Return these as a list of strings in Python.* Five images were generated per
 1101 prompt and analyzed using our custom RA-SAE model (see Appendix B), which ranked them by
 1102 the intensity with which the desired concept appeared. All images were then manually reviewed to
 1103 determine whether the blindspot was successfully depicted.

1104 As seen in the following examples of suppressed concepts, while some aspects of the target concept
 1105 occasionally appeared (e.g., a holder or string for the bird feeder blindspot and a round hole for
 1106 the glossy DVD disc blindspot), the models generally failed to generate the full concept. This
 1107 aligns broadly with our method’s assessment and supports the validity of the stress test.

1108 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Examples of images generated with a variety of prompts involving the bird feeder concept as part of the stress testing. In Kandinsky, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, SD 1.5, in which this concept was not identified as a blindspot, is able to generate it—illustrating how the method’s predictions track behavior across distinct models.



Figure 22: Examples of images generated with a variety of prompts involving the glossy DVD disc concept as part of the stress testing. In SD 1.5, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, Kandinsky, in which this concept was not identified as a blindspot, is able to generate it—again showing that the method’s predictions generalize across models.

1110 H Additional Examples: Synthesized Images



Figure 23: Additional image-caption pair examples from LAION-5B, with matching images generated from the same prompt by the diverse set of models—SD 1.5, SD 2.1, PixArt, and Kandinsky.

1111 I Concentration bounds for δ

1112 In our experiments, we estimate $\delta(k)$ using $n = 10,000$ paired samples for each concept. While this
 1113 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates in
 1114 general. To address this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ using
 1115 McDiarmid’s inequality [142]. The resulting bound is tight and demonstrates that, broadly, even with
 1116 relatively few samples we can obtain fast and accurate estimates of concept bias—a guarantee that
 1117 holds uniformly across concepts and models.

1118 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1119 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1120 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1121 *Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies*

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1122 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1123 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1124 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1125 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1126 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1127 which proves the claim. □

1128 Practically, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1129 a few reaching values up to 10. The concentration bound shows that, even for the largest observed
 1130 biases, a sample size of $n = 10,000$ yields estimates of $\hat{\delta}_n(k)$ that deviate from the true value by no
 1131 more than a small ε with high probability. This justifies our sampling strategy and confirms, more
 1132 generally, that accurate bias measurements are attainable with limited data across the full range of
 1133 concepts.

1134 J Monotonicity and Calibration-Free Interpretation of δ

1135 Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias, in
 1136 a way that is comparable across concepts and models. In practice, we use the score $\delta(k)$ for this
 1137 purpose. However, one may wonder whether such a score introduces distortions relative to more
 1138 direct quantities such as the energy gap or the odds ratio. The following result establishes that $\delta(k)$ is
 1139 a strictly increasing reparameterization of both, and therefore inherits their ordering. This guarantees,
 1140 in full generality, that no calibration is needed when using $\delta(k)$ to rank concepts.

1141 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1142 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1143 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1144 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1145 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1146 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1147 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1148 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1149 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1150 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1151 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1152 concepts.

1153

□

1154 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by conceptual
 1155 generation odds ρ_k . No calibration is necessary, and all three quantities preserve the same total
 1156 ordering over concepts—a property that holds universally, independent of the concept or model
 1157 considered.

K Stability of FID Under SAE Embeddings

In this section we establish a general, quantitative relationship between the Fréchet Inception Distance (FID) computed in the original activation space (of dimension d) and the FID after applying a (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume that D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We start by recalling a simple fact: if D is not overcomplete, orthogonal and $k = d$, then we have an isometry between A and Z , implying that the FID is perfectly preserved. However, since this case is not realistic, we then turn to the more general overcomplete case and show that FID can be bounded by the extreme singular values of D . We work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is simply $\mathcal{W}_2^2$ specialised to Gaussians.

For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically one usually finds $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof holds generally and does not rely on that. We begin with a simple lemma in the special case where D is not overcomplete.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$

This case, however, is quite unrealistic, as SAEs generally rely on overcompleteness to extract meaningful and interpretable concepts. In the more general overcomplete case, D is no longer orthonormal, but we can still have a column-orthonormal dictionary. We use this to derive a bound in terms of the extremal singular values of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1192 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1193 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1194 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1195 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1196 finishes the proof. $\square$

1197 Essentially, theorem 6 tells us, in general, that applying a column-orthogonal overcomplete SAE
 1198 dictionary cannot distort Fréchet Inception Distance by more than the square of its extremal singular
 1199 values. When D is nearly orthogonal – empirically we usually found that $\sigma_{\min}, \sigma_{\max} \approx 1$ – the result
 1200 implies that FID measured in the SAE feature space remains essentially close to the canonical FID,
 1201 across settings. $\square$

1202